$\frac{26}{30}$

FACULTY OF ENGINEERING TECHNOLOGY

Subject: Comm.& Data Transmission — Mid Exam Summer 22/23

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10:00- 8:30 د. أمجد الهندي

Q1:(Please write your answers into table only, otherwise your answers will be neglected)

Question	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*
Answer	A	A	C	B	A	C	B	A	B	D

5) B , 6) B

+16

1. Changing TV channels by using remote control, it is an example of:
   a) simplex point-to-point communication
   b) half-duplex point-to-point communication
   c) simplex multipoint communication
   d) half-duplex multipoint communication

2. Voice broadcasting mesh network contains 5 stations, where the traffic moves from 1to 2, 2 to 3 ,3 to 4 & so on m the number of links in this network are:

   a) 10 b) 5 c)24 d) 20

3. Network topology refers to the layout of:
   a) physically connected devices
   b) logically connected devices
   c) physically & logically connected devices
   d) physically & semi-logically connected devices

4. Multiple connection of devices in a network can be found in:
   a) mesh & star topologies b) bus only
   c) mesh & ring topologies d) bus & star topologies

5. End to end logic connection between application layers is done by:
   a) directly between application layers at both sides
   b) all layers
   c) directly between application & network layers at both sides
   d) logically between application & network layers at both sides

6. Power companies send simple sine wave with a frequency of 60 or 50 Hz, this wave carries :

a) power & energy
b) energy
c) power
d) nothing

7. The actual measurement of the speed of the link is:
a) bandwidth b) throughput c) data rate d) range

8. The wavelength of 600THz green light ray in air is:
a) 0.5 µm b) 0.05 µm c)0.005 µm d) 500 µm

9. Radio signals are made up to:
a) voltages & currents b) electric & magnetic fields
c) electrons & protons d) noise & data

10. The capacity of a channel with 2MHz bandwidth & SNR=3981, is:

a) 10.4 Mbps b) 24 Mbps c) 10.3 Mbps d) 23.9 Mbps

Q2: An amplifier rated at 72 w output is connected to an 8 Ω speaker.
(a) what input power is required for full power if power gain is 30dB

• $P_{out} = 72$ Watt • $R = 8\,\Omega$

• $\text{Gain} = 10 \log_{10}\left(\frac{P_{out}}{P_{in}}\right) \Rightarrow \frac{P_{out}}{P_{in}} = 10^{\left(\frac{\text{Gain}}{10}\right)}$

$\frac{P_{out}}{P_{in}} = 1000 \Rightarrow P_{in} = 0.072$ watt #

(b) what is the input voltage for rated output if the amplifier voltage gain is 40dB

$V_{out} \Rightarrow P_{out} = \frac{(V_{out})^2}{R}$

$V_{out} = \sqrt{R * P_{out}} = \sqrt{8 * 72}$

$V_{out} = 24\ V$

• $\text{Gain} = 20 \log_{10}\left(\frac{V_{out}}{V_{in}}\right)$

$\overset{2}{\cancel{\frac{40}{20}}} = \cancel{\frac{20}{20}} \log_{10}\left(\frac{24}{V_{in}}\right)$

$2 = \log_{10}\left(\frac{24}{V_{in}}\right)$

$10^2 = \frac{24}{V_{in}}$

$V_{in} = \frac{24}{10^2} \Rightarrow V_{in} = 0.24\ V$ #

X10